

Shahriyar Shawon

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Education

Drexel University

Bachelor of Science in Computer Science

Philadelphia, PA

September 2020 – June 2025

Concentrations: Operating Systems and Computational Imaging

Relevant Coursework: Operating Systems (Linux and Inferno), Concurrent Programming, Computational Photography, Computer Graphics, Machine Learning

Technical Skills

Languages/Tools: Python, C#, C, C++, Rust, Java, MySQL, PostgreSQL, OracleDB, Javascript, Git, Gitlab CI/CD

Frameworks: FastAPI, Django, Flask, .NET Core/Framework, Entity Framework, JQuery

Platforms: Linux, Windows, Docker, RP2040, Raspberry Pi, Arduino

Experience

Full-Stack Web Developer Co-Op - SFG

Bala Cynwyd, PA

Susquehanna International Group | C#, .NET, OracleDB, MySQL, WPF

September 2023 – March 2024

- Developed an API to aggregate data from 15 databases to enhance accuracy for stock information query
- Redesigned web app to use central API to simplify data sourcing
- Introduced caching mechanism by front loading requests to reduce lookup speed of stock information by 95%
- Designed and implemented role based security access system to protect privileged data

Back-End Software Developer Co-Op - MDF Metadata

Bala Cynwyd, PA

Susquehanna International Group | Python, Django, MySQL, PostgreSQL, Linux

September 2022 – March 2023

- Developed an alerting system to verify files were being received in correct amounts, to the right location, on time
- Simplified creating alerts by developing interface to make new alerts throughout the day
- Enhanced accuracy of tool for identifying users of internal files by using multiple identification methods
- Adapted old MySQL database tables to new architecture to migrate to PostgreSQL

Full-Stack Web Developer Co-Op - CorpSys

Bala Cynwyd, PA

Susquehanna International Group | C#, SQL, .NET Framework/Core

September 2021 – August 2022

- Enriched internal project management tool by gathering up to date information about developers, stakeholders, testing pipelines, environments, and code location
- Created front-end for API to allow general purpose alerts to be created
- Reduced number of bridge tables in MySQL by using Entity Framework ORM to make queries

Projects

Dynamic Image Compression Engine | C, Python, RP2040

- Built firmware for 2 RP2040 chips to interface with FPGA, IO Modules, and Power Monitor Chips
- Implemented dynamic image format to store lossy and lossless blocks of data using QOI and PNG codecs
- Created desktop app to control camera settings on the fly and log power/data metrics from hardware
- Engineered pipeline to compare software based encoding performance to hardware based encoding

Chip 8 Emulator | Rust

- Modeled Chip 8 CPU as data structure to maintain internal state
- Implemented Fetch-Decode-Execute loop on Chip 8 ROMs
- Created disassembler to assist in emulator debugging
- Used SDL to visualize video memory contents

Drexel Course Information API | FastAPI, grequests, BeautifulSoup

- Asynchronously scraped course information by parsing information from 8200+ Drexel Courses across 175 subjects and 15 departments and stored in SQLite database
- Created API to be able to query for classes based on school, subject, credits, and name
- Enabled users to see the all potential prerequisite paths a student could take to take a class, and see all the classes that require one class as a prerequisite